

ANIRUDH VERMA

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EDUCATION

Indian Institute of Science, Bengaluru (IISc) Physics Major, Bachelor and Master of Science (Research). CGPA: 8.5/10 (BS), 8.4/10 (BS - MS)	2020 – 2025
Holy Wisdom School, Champawat, Uttarakhand High School (AISSCE): 97.6%	2020
Udayan International School, Champawat, Uttarakhand Middle School (AISSE): 94.2%	2018

PUBLICATIONS

A Quantum-Walk Representation of Color-Ordered MHV Scattering Amplitudes Anirudh Verma, C. M. Chandrashekar (Under review at <i>Physical Review D</i>): arXiv:2607.02456	2026
Photonic Simulation of Beyond-Quantum Nonlocal Correlations (e.g. Popescu-Rohrlich Box) with Non-Signaling Quantum Resources Kunal Shukla, Anirudh Verma, Kanad Sengupta, Sanchari Chakraborti, Manik Banik, C. M. Chandrashekar (Under review at <i>NPJ Quantum Information</i>): arXiv:2509.26271	2025
Four-Qubit Photonic System for Publicly Verifiable Quantum Random Numbers and Generation of Public and Private Key Mayalakshmi Kolangatt, Anirudh Verma, Sujai Matta, Kanad Sengupta, C. M. Chandrashekar <i>Physical Review A</i> 110, 032615	2024

BACHELOR AND MASTER THESIS

Entanglement Certified Quantum Random Number Generation using Polarisation Entangled Photons Supervisor: Prof. C. M. Chandrashekar, Associate Professor, IISc Bangalore (Bachelor Thesis Advisor)	2023–24
Response and Correlations of Weakly Measured Free Fermions Supervisor: Prof. Sumilan Banerjee, Associate Professor, IISc Bangalore (Master Thesis Advisor)	2024–25

RESEARCH PROJECTS

Quantum Walk Dynamics and PR Correlations (Project Assistant) Supervisor: Prof. C. M. Chandrashekar, Associate Professor, IISc Bangalore - Currently working on quantum tokenized signature scheme. - Simulated post-quantum nonlocal correlations, such as Popescu-Rohrlich boxes, using intrinsically non-signaling resources implemented on a four-qubit photonic platform. - Simulated quantum walks in spin Hamiltonians (e.g., XXZ, TFIM) to understand the dynamics in a lattice configuration. - Investigated scattering amplitudes using quantum walks.	June 2025 – Present
Quantum Simulation using Variational and Sample-Based Methods Supervisor: Prof. Viki Kumar Prasad, University of Calgary - Built a quantum chemistry simulation pipeline using PySCF and Qiskit Nature with ACP-corrected Hamiltonians for noncovalent systems. - Performed active space reduction and integral projection to enable efficient quantum circuit simulation. - Applied VQE with UCCSD ansatz and benchmarked against SCF and exact diagonalization. - Implemented Sample-Based Quantum Diagonalization (SQD) using subspace projection from parameterized quantum states.	Sep 2025 – Dec 2025
Linear Response Theory across Measurement-Induced Entanglement Transition Supervisor: Prof. Sumilan Banerjee, Associate Professor, Department of Physics, IISc Bangalore - Worked in the field of open quantum many-body systems, studying measurement-induced entanglement transitions in free fermionic chains under continuous weak monitoring. - Used stochastic Schrödinger equations to simulate quantum trajectories and explore the interplay of unitary dynamics with local number measurements. - Benchmarked a crossover from logarithmic to area-law entanglement scaling and analyzed density-density and	Sep 2024 – Apr 2025

current-current correlation decay.

- **Term Paper:** *Simulation of Quantum Trajectories and Measurement-Induced Entanglement Transition*

Quantum Random Number Generation using Polarisation Entangled Photons

Sep 2023 – Apr 2024

Supervisor: Prof. C. M. Chandrashekar, Associate Professor, IISc Bangalore

- Developed a **polarisation-entangled photon source** via **Type-II SPDC** using a PPKTP crystal, achieving high-fidelity entanglement verified through **Bell-CHSH inequality** and **quantum state tomography**.
- Demonstrated protocols for **publicly verifiable QRNG**, **entanglement verification**, and simultaneous generation of **public/private cryptographic keys**.
- Generated **secure random bits at 185 kbps**, with outputs passing all **15 NIST randomness tests** without post-processing.

Quantum Optics: Mondal Lab

Aug 2023 – Oct 2023

Supervisor: Prof. Partha Pratim Mondal, Professor, IISc Bangalore

- Studied super-resolution optical microscopy, specifically **Single Molecule Localisation Microscopy**, and methods for improving spatial resolution beyond diffraction-limited techniques.

Diagnostics, Control and Data Processing for the STOR-M Tokamak Experiments

May 2023 – July 2023

Supervisor: Prof. Chijin Xiao, Professor, University of Saskatchewan, Canada

- Developed an interactive data-analysis code for STOR-M Tokamak plasma discharges across different oscilloscopes; modified legacy MDSPlus archiving programs (MATLAB) and extended a Python GUI for fluctuation-signal analysis.

MACHINE LEARNING PROJECTS

Self-directed projects building applied experience across supervised, unsupervised, deep learning, and reinforcement learning.

Loan Approval Prediction : End-to-End Supervised ML Pipeline

2026

Personal project

- Built a complete binary classification pipeline to predict loan approval, covering exploratory data analysis, feature engineering, model training, and evaluation.
- Implemented and compared **K-Nearest Neighbors**, **Logistic Regression**, and **Naive Bayes** classifiers, evaluating performance using standard classification metrics (accuracy, precision, recall, F1, ROC-AUC).

RELEVANT COURSES

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| • Classical and Quantum Optics | • Introduction to Quantum Computation |
| • Condensed Matter Physics I | • Quantum Mechanics II |
| • Non-Linear and Ultrafast Optics | • Analysis and Linear Algebra II |
| • Discrete Photonics and Quantum Analogies | • Statistical Mechanics |
| • Algorithms and Programming | • Electromagnetic Theory |
| • Introduction to Fourier Optics | • Non-Equilibrium Quantum Many-Body Dynamics |
| • Introduction to Quantum Measurement | |

TECHNICAL SKILLS

- **Programming:** Python, SQL, MATLAB, Mathematica, HTML, CSS
- **Libraries:** NumPy, SciPy, SymPy, Pandas, Scikit-learn, PyTorch, Qiskit, QuTiP, QuSpin, TeNPy
- **Software:** L^AT_EX, SciDAVis, WordPress
- **Operating Systems:** Linux, Windows
- **Languages:** English, Hindi

AWARDS AND HONORS

- Mitacs* Globalink Research Intern, 2023
- Honoured by the District Magistrate for securing 2nd rank in AISSCE 2019–20 at the district level
- Qualified JEE Mains & JEE Advanced in the top 1 percentile, 2020
- IISc Promotional Scheme Scholar, Nov 2020 – Apr 2025

- IBM Quantum Challenge Badge, 2024
- IBM Quantum Excellence Badge, 2025

*Mitacs Globalink Research Internship is a competitive initiative for international undergraduates to participate in a 12-week research internship under the supervision of Canadian university faculty members across science, engineering, mathematics, and the humanities.

WORKSHOPS AND CONFERENCES

Quantum India, Bangalore Presented published work on four-qubit systems.	2025
A Hundred Years of Quantum Mechanics — ICTS Bangalore Attended and participated in discussions on the foundations and future of quantum mechanics.	2025
TQC 2025 — IISc Bangalore Volunteered, chaired sessions across all days, and developed the official conference website TQC2025 .	2025
Other Workshops IBM–IISc Workshop on Quantum Computing and Hybrid Clouds ; The Microsoft Garage Student Innovation Workshop , Workshop 1 & 2	2022

EXTRACURRICULAR AND VOLUNTEER EXPERIENCE

- **Co-Convener, Carrom Club, IISc** (Nov 2021 – July 2025): Conducted offline events and inter-departmental competitions.
- **Coordinator, Pravega Innovation Summit** (Sep 2022): Organized an innovation competition for registered startups with judges and speakers.
- **Volunteer, Notebook Drive** (2022): Led the stationery-distribution initiative.
- Represented IISc in inter-college and university football and carrom competitions (2022–24).

REFERENCES

Prof. Sumilan Banerjee

Associate Professor
Department of Physics
Indian Institute of Science, Bengaluru
sumilan@iisc.ac.in

Prof. C. M. Chandrashekar

Associate Professor
Department of Electronic Systems Engineering
Indian Institute of Science, Bengaluru
chandrasm@iisc.ac.in